

A Graph Diffusion Model for Environmental Justice Pollution Exposure in Chelsea and Everett, Massachusetts

Abstract

Chelsea and Everett, Massachusetts are small, dense, highly connected urban communities in the Lower Mystic region. Their neighborhoods sit near major roads, freight corridors, industrial parcels, schools, and regional infrastructure. This paper develops a graph-based model for estimating transportation-related pollution exposure and for comparing policy interventions that could reduce unequal environmental burdens. The central modeling idea is to represent the study area as a spatial influence graph: road segments, block groups, schools, and industrial facilities become nodes, while nearby locations are connected by distance-decay edges. Pollution source strength is assigned to roads and facilities using estimated traffic intensity, truck share, and facility-emission scores. Exposure then propagates through the network using a steady-state graph diffusion equation. The output is not a regulatory air-quality prediction in physical units such as micrograms per cubic meter. Instead, it is a relative exposure score that allows different interventions to be compared within the same model.

The final run used 486 nodes, including 370 road nodes, 69 block-group receptor nodes, 42 school nodes, and 5 facility nodes, with 20,303 spatial edges. Baseline exposure was higher for the modeled environmental justice group than for the non-environmental-justice comparison group. Four interventions were tested: anti-idling enforcement near schools, electric freight priority on high-source roads, a low-emission zone, and industrial source reduction. Anti-idling near schools produced the largest environmental justice exposure reduction in the current run, narrowly ahead of electric freight priority. A sensitivity analysis over diffusion strength, connection radius, and decay distance showed that the main road-based recommendations are generally robust, although electric freight priority becomes stronger when diffusion is allowed to travel farther through the graph. The results suggest that city policy should prioritize school-adjacent freight and idling controls, while pairing them with longer-term electrification of high-source freight corridors and more detailed collection of real traffic, truck, facility, wind, and sensor data. Because several inputs were estimated rather than directly measured, the results should be interpreted as a first-pass decision-support model rather than a final regulatory analysis.

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1 Introduction

Air pollution is not distributed evenly across cities. It is shaped by where roads are built, where freight moves, where industrial land uses are placed, where housing is located, and which neighborhoods have the least political and economic power to avoid environmental burdens. Chelsea and Everett, Massachusetts provide a compact but important case study of this problem. Both communities are located close to major regional transportation corridors and industrial activity in the Lower Mystic area. Their residents also live in dense neighborhoods where schools, homes, truck routes, and commercial corridors are often close to one another. In this setting, the practical question is not simply whether air pollution exists. The more useful question is: where should limited municipal attention be directed first if the goal is to reduce unequal pollution exposure?

This paper answers that question with a mathematical model. The project uses a graph diffusion framework to estimate relative exposure and compare candidate policy interventions. The graph model represents the urban environment as a network of places. A census block group is a node because it represents a residential population. A road segment is a node because it can act as a source of traffic-related emissions. A school is a node because it is a sensitive receptor where children spend time. An industrial or freight facility is a node because it can represent stationary emissions, truck generation, idling, loading, or industrial activity. Edges connect nearby nodes, and their weights represent spatial influence. A road that is physically close to a block group or school has a stronger edge than a faraway road. The model then estimates how source strength diffuses through the weighted graph.

The value of this approach is that it connects mathematical modeling to policy. A city does not only need to know where pollution may be high; it needs to know which action would produce the greatest improvement. In this project, four interventions are simulated. Anti-idling enforcement near schools reduces road-source strength within 300 meters of school nodes. Electric freight priority reduces source strength for the highest-source road nodes. A low-emission zone reduces road and facility sources near the center of the study area. Industrial source reduction reduces source strength from facility nodes. Each intervention changes the source vector, the diffusion model is rerun, and exposure reduction is measured for all block groups, environmental justice block groups, and schools.

The model is intentionally transparent, and it does not attempt to replace atmospheric chemistry models, field monitoring, or public-health analysis. It also does not claim to prove causation between a specific road and a specific health outcome. Instead, it asks a narrower and more suitable modeling question: if traffic- and facility-related source strengths are represented on a spatial graph, which policy scenario most reduces modeled exposure for vulnerable communities? This is the kind of question that can be answered with graph theory and optimization even when perfect air-quality data are not available.

2 Why the Problem Matters

Transportation-related air pollution is important because people encounter it at the scale of daily life: children walk near roads to school, families live beside bus routes, arterials, and intersections, workers travel along congested freight corridors, and trucks idle near loading areas. Industrial land uses can concentrate in places that are close to lower-income residential areas. Even if regional air quality is improving, exposure can remain locally unequal, meaning a community can have acceptable average conditions while certain blocks or schools remain much more burdened than others.

Chelsea and Everett are useful for a modeling study because the geography is dense and policy-relevant. In a sprawling region, a pollution model may require many assumptions about long-distance dispersion, commuting, land-use transitions, and regional meteorology. In a compact urban-industrial area, the first-order relationships are easier to state: major roads, truck corridors, schools, industrial parcels, and residential block

groups are all close enough that local spatial structure matters. CTPS's Route 16 Priority Corridor Study describes the corridor through Chelsea and Everett as an area with high traffic and truck activity, and it reports daily traffic volumes in the range of 38,000 to 70,000 vehicles per day on Route 16, with daily truck shares between 3 percent and 5 percent and a much higher truck concentration on Second Street (Central Transportation Planning Staff, 2019). Those numbers justify a model that treats road and freight activity as central sources.

The environmental justice dimension is also central. MassGIS provides a 2020 Environmental Justice Populations layer at the Census 2020 block group level, intended to identify environmental justice populations across Massachusetts (MassGIS, 2021). Such a layer matters because the goal of the project is not merely to reduce total exposure in the abstract. A policy that reduces exposure in an already less-burdened area may look good in total, but it may not improve equity. Conversely, a policy with a smaller total effect may be better if it helps high-burden neighborhoods, children, or dense residential areas. This project therefore evaluates policies using an environmental justice benefit metric, not only a total exposure metric.

The public-health motivation is broad, but the model itself is cautious. Pollution exposure can be associated with asthma and other respiratory risks, and traffic-related pollutants such as nitrogen oxides, fine particles, diesel particulate matter, and black carbon are common concerns near roadways and freight activity. However, this paper does not estimate disease outcomes because health outcomes depend on many factors, and a student-level graph diffusion model should not overclaim. Instead, the model focuses on relative exposure reduction by asking which intervention reduces modeled exposure more than another, making the result useful as a screening and prioritization tool for deciding where more monitoring, enforcement, or planning analysis should occur.

3 Background and Related Data Context

The project connects three domains: environmental justice, urban freight exposure, and graph diffusion. Environmental justice analysis asks whether environmental burdens are distributed unequally across communities. Urban freight exposure asks how trucks, industrial sites, highways, ports, warehouses, and arterial roads affect nearby residents. Graph diffusion provides a mathematical way to model how a quantity spreads or propagates over a network.

Several public data sources support this kind of analysis. MassGIS publishes a 2020 environmental justice population layer, named EJ_POLY, and notes that it shows which Census 2020 block groups are classified as environmental justice populations (MassGIS, 2021). MassGIS also publishes the MassGIS–MassDOT Roads layer, described as the official state-maintained street transportation dataset available from MassGIS (MassGIS, 2025). The MassGIS Massachusetts Schools point layer provides the locations of pre-kindergarten through high school facilities across the state (MassGIS, 2025). These layers allow a model to define receptor nodes, source nodes, and sensitive receptor nodes.

For traffic magnitudes, the CTPS Route 16 Priority Corridor Study is especially relevant because it is specific to Chelsea and Everett. CTPS reports that Route 16 weekday traffic volumes range from about 38,000 to 70,000 vehicles per day and that Route 16 truck traffic ranges between 3 percent and 5 percent; it also reports that Second Street carries more than 1,600 trucks daily, about 12 percent of its traffic (Central Transportation Planning Staff, 2019). Those values were used to calibrate reasonable ranges for imputed traffic and truck-share values when direct road-level count data were not available for every road node.

For facility sources, the EPA National Emissions Inventory (NEI) provides the relevant conceptual category. EPA describes the NEI as a comprehensive estimate of emissions of criteria pollutants, criteria precursors, and hazardous air pollutants. It also describes point sources as larger stationary sources such as industrial

facilities, electric power plants, airports, and smaller industrial, non-industrial, and commercial facilities (U.S. Environmental Protection Agency, 2026a). The 2020 NEI includes criteria air pollutants and precursors, hazardous air pollutants, and some greenhouse gas data for facilities nationwide and on a countywide basis (U.S. Environmental Protection Agency, 2026b). In this project, facility nodes are represented with relative emission scores rather than official pollutant-specific tonnage values. That is an important limitation, but it is consistent with the screening nature of the model.

Air-quality sensors could improve future validation because the iSUPER project in Chelsea uses QuantAQ MODULAIR sensors to measure particulate matter and gases including NO_x, ozone, and carbon monoxide, with some sensors also measuring wind speed and wind direction (iSUPER, 2026). Sensor data are not required for the current graph diffusion run, but they would be valuable for checking whether predicted exposure hotspots align with measured pollution patterns. In a future version of this project, observed PM, NO_x, or ozone values could be joined to the closest sensor nodes and used for calibration or validation.

4 Data Used in the Current Model

The current model run used a cleaned, filtered dataset representing Chelsea and Everett with a one-kilometer study-area buffer. The final graph contained 486 nodes and 20,303 edges. The node set included 370 roads, 69 census block groups, 42 schools, and 5 industrial or freight-related facility nodes. Table 1 summarizes the node inventory, and Figure 1 visualizes it.

Table 1: Node inventory in the final graph.

Node Type	Count
Road	370
Block Group	69
School	42
Facility	5

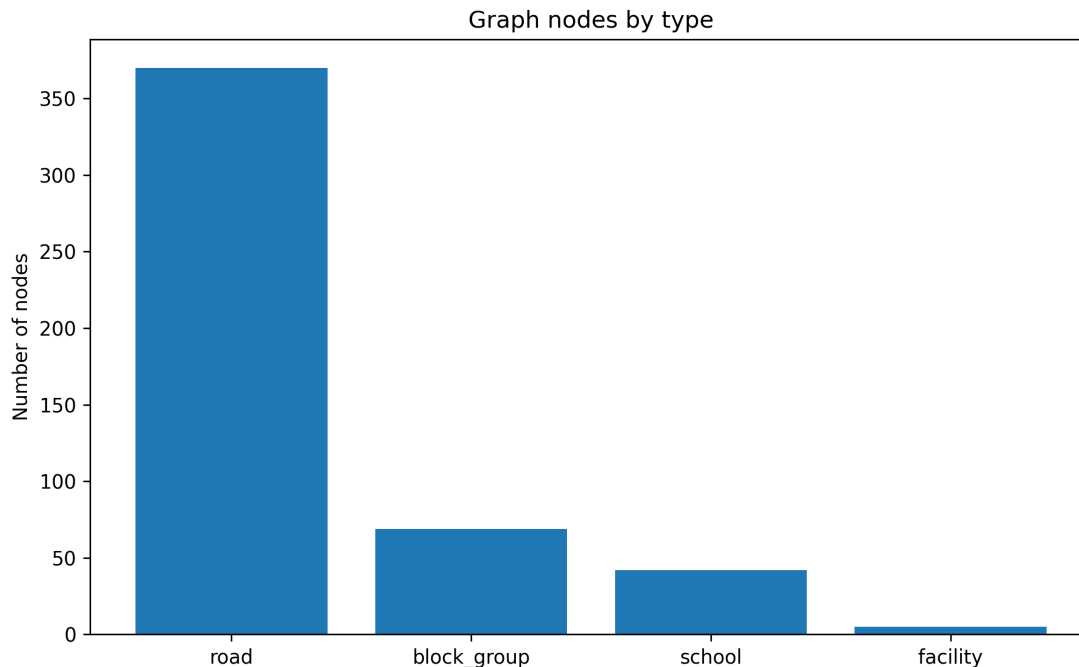


Figure 1: The model is dominated by road nodes, which is expected for a transportation-related pollution study. Block groups and schools serve primarily as receptor nodes, while the smaller facility layer represents stationary or freight-related sources.

The block-group layer represented the residential exposure units. Each block group carried a population value and a binary environmental justice flag. The current run included 34 environmental justice block groups with a combined population of 43,509 and 35 non-environmental-justice comparison block groups with a combined population of 46,331. The comparison group was necessary because the previous run had labeled all block groups as environmental justice areas, which made the equity comparison impossible. For this run, the environmental justice and non-environmental-justice split should be treated as a modeling approximation rather than a final official designation. In a final city-facing version, this field should be replaced by the official MassGIS environmental justice attributes.

The road layer provided the main pollution-source network. Each road node had an estimated annual average daily traffic value, an estimated truck percentage, and a derived source strength. Because complete measured road-level counts were not available for all nodes, traffic values were imputed using a gravity-style scaling procedure. The purpose of the imputation was not to invent official traffic counts; it was to create a plausible range of heterogeneous road-source values so that the model could distinguish high-source roads from lower-source roads. The scaling used the preliminary model exposure score as a proxy for spatial centrality and traffic influence. The estimated AADT values ranged from 10,000 to 60,000 vehicles per day, and estimated truck shares ranged from 3 percent to 8 percent. These ranges were chosen to be consistent with the magnitude of regional traffic and truck values reported in the CTPS Route 16 study, while still allowing local roads to have lower values than Route 16.

The facility layer added five nodes representing likely industrial or freight-related sources. These included large or regionally important facilities and freight-relevant industrial areas. Facility nodes were assigned emission scores on a zero-to-one scale. This produced a stationary-source layer, but it was much smaller than the road layer. As a result, industrial source reduction had a weaker effect than road-based policies in the current run. That does not prove industrial sources are unimportant in reality; it only means that, in this version

of the graph, their modeled source contribution was smaller than the combined road network.

School nodes were included as sensitive receptors because they do not emit pollution in the model, but they still receive exposure from nearby source nodes and represent places where children may be vulnerable. The anti-idling intervention used school locations directly by reducing road-source strength for roads within 300 meters of school nodes. This represents a policy in which enforcement or idling reduction is concentrated where children may be exposed during arrival, dismissal, and outdoor activity periods.

5 Modeling Assumptions

Before defining the graph diffusion equation, it is important to state the assumptions that make the model possible. These assumptions do not make the model “wrong.” They define the level of abstraction. The project is designed as a screening and prioritization model, not as a regulatory air-dispersion model. Its purpose is to compare interventions consistently under a shared set of rules.

Key Assumptions

The exposure score in this paper is a relative index. It should be interpreted as a modeled burden score inside this graph, not as a measured PM_{2.5}, NO₂, diesel particulate matter, or respiratory-risk concentration.

First, the model assumes that local pollution influence can be approximated by spatial proximity. In the real world, pollution movement depends on wind, atmospheric stability, building geometry, street canyons, fleet mix, vehicle speed, congestion, idling, and chemical transformation. The current model does not explicitly simulate those processes. Instead, it uses an exponential distance-decay edge weight. This assumes that nearby sources matter more than distant sources and that influence decreases smoothly with distance. This is a reasonable first-pass assumption for comparing local interventions, but it is not a substitute for atmospheric dispersion modeling.

Second, the model assumes a steady-state exposure process. The source vector represents a typical relative burden rather than a specific hour of the day. This means the model does not distinguish between morning truck activity, afternoon school pickup, night delivery, weekend traffic, or seasonal patterns. In practice, the anti-idling scenario may be especially important during school arrival and dismissal, while freight electrification may matter more across the entire day. The present model averages those patterns into one static network so that interventions can be compared simply.

Third, road-source strength is treated as a function of traffic volume and truck share. Because direct measured traffic counts were not available for every road node, the current run used imputed AADT and truck-share values. These were scaled to plausible ranges using the CTPS Route 16 corridor values as a guide. In other words, the model uses traffic estimates to create meaningful variation among roads, but the values should not be described as official traffic counts. A final city-facing version should replace them with MassDOT, CTPS, or local count data wherever possible.

Fourth, the environmental justice flag is used as a binary modeling field. The model compares population-weighted exposure in block groups labeled as environmental justice areas to exposure in comparison block groups. In reality, environmental justice status can be based on different criteria, including income, race or ethnicity, English isolation, and combinations of indicators. A more detailed version should preserve those categories instead of reducing them to one binary variable.

Fifth, policy effects are represented as percentage reductions in source strength. Anti-idling near schools is modeled as a 25 percent reduction for nearby road nodes, electric freight priority as a 40 percent reduction for the highest-source road nodes, the low-emission zone as a 35 percent reduction inside the zone, and industrial source reduction as a 30 percent reduction for facility nodes. These are scenario assumptions, not measured

outcomes. They are useful because they allow policies to be compared on the same scale, but the actual effectiveness of each policy would depend on enforcement, compliance, fleet turnover, infrastructure, and funding.

Sixth, the facility layer is simplified. Five facility nodes are included, each with a relative emission score. This gives the model a stationary-source component, but it almost certainly underrepresents the full industrial and freight system around Chelsea and Everett. More complete EPA NEI, permitting, warehouse, terminal, loading-zone, rail, and marine data could increase the role of non-road sources.

Figure 2 summarizes the overall workflow. The important point is that assumptions enter the model before the diffusion equation is solved: the data determine the nodes, the assumed spatial decay determines the edges, and the assumed policy effects determine how the source vector changes.

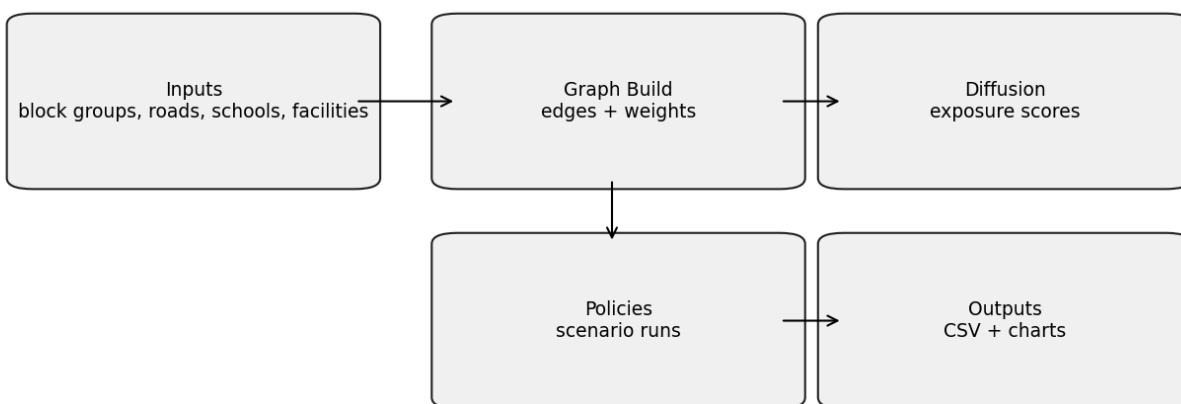


Figure 2: Workflow of the graph diffusion model. Raw spatial inputs are converted into nodes, spatial influence edges are built, the diffusion equation generates exposure scores, policy scenarios modify source strengths, and outputs are exported as tables and charts.

6 Main Model

The modeling framework has three connected parts: a source-strength model, a spatial graph model, and an evaluation model. The source-strength model assigns relative pollution intensity to roads and facilities. The spatial graph model determines how strongly each source can influence nearby homes, schools, and other receptors. The evaluation model then summarizes the resulting exposure scores for the whole study area, environmental justice block groups, and school nodes. Keeping these pieces separate is important because a city can improve traffic data, facility data, or equity weights later without rewriting the whole mathematical structure.

6.1 Graph Representation

The model is built as a steady-state graph diffusion model. Let $G = (V, E)$ be a weighted graph where V is the set of nodes and E is the set of spatial influence edges. Nodes can represent roads, block groups, schools, facilities, or sensors. An edge is not a travel route; instead, it is a mathematical link indicating that two nearby places may influence one another through local pollution exposure. This distinction matters because the model

is not trying to route vehicles through a street network. It is estimating how source intensity near one location can affect nearby receptors.

Each edge (i, j) has a nonnegative weight w_{ij} , which is larger when nodes are closer and smaller when nodes are farther apart. In the current implementation, edges are created between nodes within a radius of 1,500 meters. The base edge weight follows exponential distance decay:

$$w_{ij} = \exp\left(-\frac{d_{ij}}{\lambda}\right),$$

where d_{ij} is the distance between nodes i and j , and $\lambda = 500$ meters is the decay distance. This formula means that nearby roads, schools, and block groups interact strongly, while farther pairs still can interact but with much weaker influence. If one endpoint is a source-like node and the other is a receptor-like node, the edge weight is increased by a factor of 1.25 so that direct source-to-receptor relationships receive slightly more importance than links between similar node types.

6.2 Diffusion Equation

After the weighted adjacency matrix W is built, it is row-normalized into a transition matrix P . If D is a diagonal matrix with $D_{ii} = \sum_j W_{ij}$, then

$$P = D^{-1}W.$$

The row-normalized matrix P describes how one unit of influence at a node is distributed across its neighbors. This prevents highly connected nodes from automatically dominating only because they have more edges; instead, their influence is divided among the nearby places to which they are connected. The exposure vector x is calculated from the source vector s , where s is positive for roads and facilities and equal to zero for block groups and schools. The steady-state equation is

$$x = s + \alpha Px,$$

which is equivalent to

$$x = (I - \alpha P)^{-1}s,$$

where I is the identity matrix and α is the diffusion strength. In this run, $\alpha = 0.65$, so the model gives meaningful weight to nearby indirect influence while still keeping direct source strength important.

The same equation can also be understood through the series expansion

$$x = s + \alpha Ps + \alpha^2 P^2 s + \alpha^3 P^3 s + \dots$$

The first term is direct source strength at each node. The second term represents one-step influence from neighboring nodes, the third term represents two-step influence through the graph, and later terms represent increasingly indirect pathways. Because $0 \leq \alpha < 1$, the later terms shrink and the steady-state solution remains stable. This interpretation is useful for policy analysis because it explains why a road can affect a block group even if the two are not directly connected: influence can move through intermediate nearby nodes, but it weakens with each step.

6.3 Source Strength Model

Road source strength was computed from estimated traffic and truck values. First, annual average daily traffic and truck percentage were normalized to scores between 0 and 1:

$$T_i = \frac{\text{AADT}_i - \min(\text{AADT})}{\max(\text{AADT}) - \min(\text{AADT})},$$

$$R_i = \frac{\text{truckPct}_i - \min(\text{truckPct})}{\max(\text{truckPct}) - \min(\text{truckPct})}.$$

The road-source score was then defined as

$$s_i = T_i(1 + 2R_i).$$

This equation gives higher source strength to roads that combine heavy traffic with a high truck share, reflecting the idea that diesel and freight activity can create larger near-road burdens than passenger traffic alone. The maximum road-source value is approximately 3 when both normalized traffic and normalized truck scores are high. Facility source strength was computed from a normalized emission score, while block groups and schools had source strength equal to zero because they are receptors rather than modeled pollution sources.

6.4 Policy Simulation Logic

Each policy scenario keeps the graph structure fixed and changes only the source vector s . This design makes the comparisons fair because the same neighborhoods, schools, roads, distances, and edge weights are used in every run. A policy therefore wins only if its source reductions occur at nodes that matter strongly for the exposure metrics. For example, reducing a high-source road far from environmental justice receptors may reduce total source strength but produce a smaller equity benefit than reducing a moderate-source road directly connected to schools and EJ block groups.

The output should be read as a relative exposure index rather than a pollutant concentration. A score of 2 is not twice a measured PM_{2.5} concentration, and a score of 0.5 is not an official air-quality standard. The score is useful because every policy is evaluated inside the same graph with the same assumptions, so differences between baseline and policy scenarios indicate which intervention is more effective within this decision-support model.

7 Policy Scenarios

Four policy scenarios were compared against the baseline so that the same graph could be used to test different ways of reducing the source vector.

7.1 Anti-Idling Near Schools

The anti-idling scenario reduced road-source strength by 25 percent for roads within 300 meters of school nodes. This scenario represents targeted enforcement near schools, school-bus loading zones, pickup/drop-off areas, and nearby streets where idling trucks or buses may contribute to localized exposure. It is a spatially targeted intervention because it does not reduce every road; it only affects roads close to sensitive receptor nodes.

7.2 Electric Freight Priority

The electric freight scenario reduced source strength by 40 percent for the top 20 percent of road nodes by source strength. This represents prioritizing high-source freight corridors for electric trucks, zero-emission delivery requirements, charging infrastructure, or freight decarbonization incentives. The updated code selects the top n roads directly rather than selecting all roads tied at a cutoff. This is important because the earlier version accidentally allowed all roads to be selected when all source strengths were equal.

7.3 Low-Emission Zone

The low-emission zone scenario reduced source strength by 35 percent for road and facility nodes within 1,200 meters of the graph center. This represents a geographically defined clean transportation zone or low-emission priority district. It is broader than anti-idling but less targeted to schools. Because it is based on location rather than source ranking, it can reduce total exposure strongly if the central zone overlaps many sources.

7.4 Industrial Source Reduction

The industrial source scenario reduced facility-node source strength by 30 percent. This represents improved stationary-source controls, cleaner industrial operations, emissions caps, or targeted permitting/enforcement actions. In the current model, this policy has a limited effect because only five facility nodes are included and the road layer dominates the source network.

8 Evaluation Metrics

The model reports several metrics so that overall reduction, population-weighted reduction, environmental justice reduction, and school reduction can be compared directly. Total block-group exposure is the sum of exposure scores over all block-group nodes:

$$E_{\text{total}} = \sum_{i \in B} x_i,$$

where B is the set of block groups. Population-weighted exposure is

$$E_{\text{pop}} = \frac{\sum_{i \in B} P_i x_i}{\sum_{i \in B} P_i},$$

where P_i is block-group population, and environmental justice population-weighted exposure is the same calculation restricted to environmental justice block groups:

$$E_{\text{EJ}} = \frac{\sum_{i \in B_{\text{EJ}}} P_i x_i}{\sum_{i \in B_{\text{EJ}}} P_i}.$$

School exposure is the average exposure over school nodes:

$$E_{\text{school}} = \frac{1}{|S|} \sum_{i \in S} x_i.$$

Policy benefit is measured as reduction from baseline:

$$\Delta E_{\text{EJ}} = E_{\text{EJ, baseline}} - E_{\text{EJ, policy}}.$$

Policies are ranked by ΔE_{EJ} , because the main research question is equity-centered. This is why a policy can rank first even if another policy has a slightly larger total block-group reduction.

9 Implementation and Reproducibility

The computational workflow was organized as a reproducible Python pipeline whose first stage loads and standardizes the raw spatial data. Block groups, roads, schools, facilities, and sensors are converted to a common projected coordinate reference system so that distances are measured in meters rather than degrees. This matters because edge weights depend directly on distance. If latitude and longitude were used without projection, a 300-meter school buffer or a 1,500-meter graph edge radius would be mathematically inconsistent. The model uses a Massachusetts projected coordinate system so that all distance-based operations occur on a local metric scale.

The second stage filters the study area by focusing on Chelsea and Everett and applying a buffer around the block-group study area. The buffer is important because pollution sources just outside a municipal boundary can still influence receptors inside the city. A road or facility does not stop mattering simply because it lies across a boundary line. At the same time, the buffer prevents the model from expanding to the entire state, which would dilute the analysis and make the graph unnecessarily large. This balance gives the model a local scale while still acknowledging boundary spillover.

The third stage creates a unified node table in which each row is a node with a node identifier, node type, source strength, receptor variables, and geometry. Roads and facilities can have positive source strengths, while block groups and schools have zero source strengths but act as receptors. This separation is important because the model distinguishes where pollution begins from where exposure is measured. A school, for example, does not generate pollution in the model, but its exposure score can still be high if it is near strong source nodes.

The fourth stage builds the graph edges by using a spatial index to identify all node pairs within the selected radius, which is more efficient than checking every possible pair manually. With 486 nodes, a complete graph would have more than 117,000 possible undirected pairs, while the radius filter reduces this to 20,303 meaningful spatial edges. The edge weight is then computed using exponential distance decay. The result is a graph that is dense enough to represent spatial influence, but not so dense that every location affects every other location equally.

The fifth stage solves the diffusion equation by building a sparse adjacency matrix, normalizing it, constructing $I - \alpha P$, and solving the linear system. Sparse matrices are important because most nodes are not connected to every other node. Solving the system directly gives the steady-state exposure vector without needing to run many iterative diffusion steps. The same edge matrix is reused for each policy scenario, while the source vector changes according to the policy intervention.

The final stage exports the results by writing a policy comparison table, a baseline node-exposure table, a block-group exposure table, a school exposure table, source contribution outputs, and edge data. This is useful because the analysis can be checked at several levels. If a policy result looks surprising, the source contribution file can identify whether one or two road nodes dominate the result. If school exposure seems high, the school table can identify which school nodes are most affected. If equity results seem unstable, the block-group table can be inspected to see whether exposure is concentrated in a few high-population receptors.

This structure makes the project reproducible and expandable. Replacing estimated traffic values with official traffic counts does not require rewriting the model; it only requires updating the input table. Adding

sensors does not require changing the diffusion equation; it adds another node type and a validation layer. Adding cost-effectiveness requires a new policy metric, not an entirely new graph. That modularity is one reason the graph approach is useful for a student research project and for early-stage city planning.

10 Results

The baseline total block-group exposure was 49.636, the baseline environmental justice population-weighted exposure was 0.940, the non-environmental-justice population-weighted exposure was 0.556, and the baseline mean school exposure was 1.157. Since the exposure score is unitless, these values should be interpreted comparatively, not as pollutant concentrations. The main result is that modeled exposure is higher for the environmental justice group, and that different policies reduce different parts of the graph.

Table 2 reports the policy comparison, and Figure 3 shows the same result visually.

Table 2: Policy comparison ranked by environmental justice exposure reduction.

Policy	Rank	Total Reduction	EJ Reduction	School Reduction
Anti Idling Near Schools	1	3.457	0.063	0.080
Electric Freight Priority	2	2.022	0.062	0.044
Low Emission Zone	3	3.135	0.047	0.033
Industrial Source Reduction	4	0.050	0.001	0.001
Baseline	5	0.000	0.000	0.000

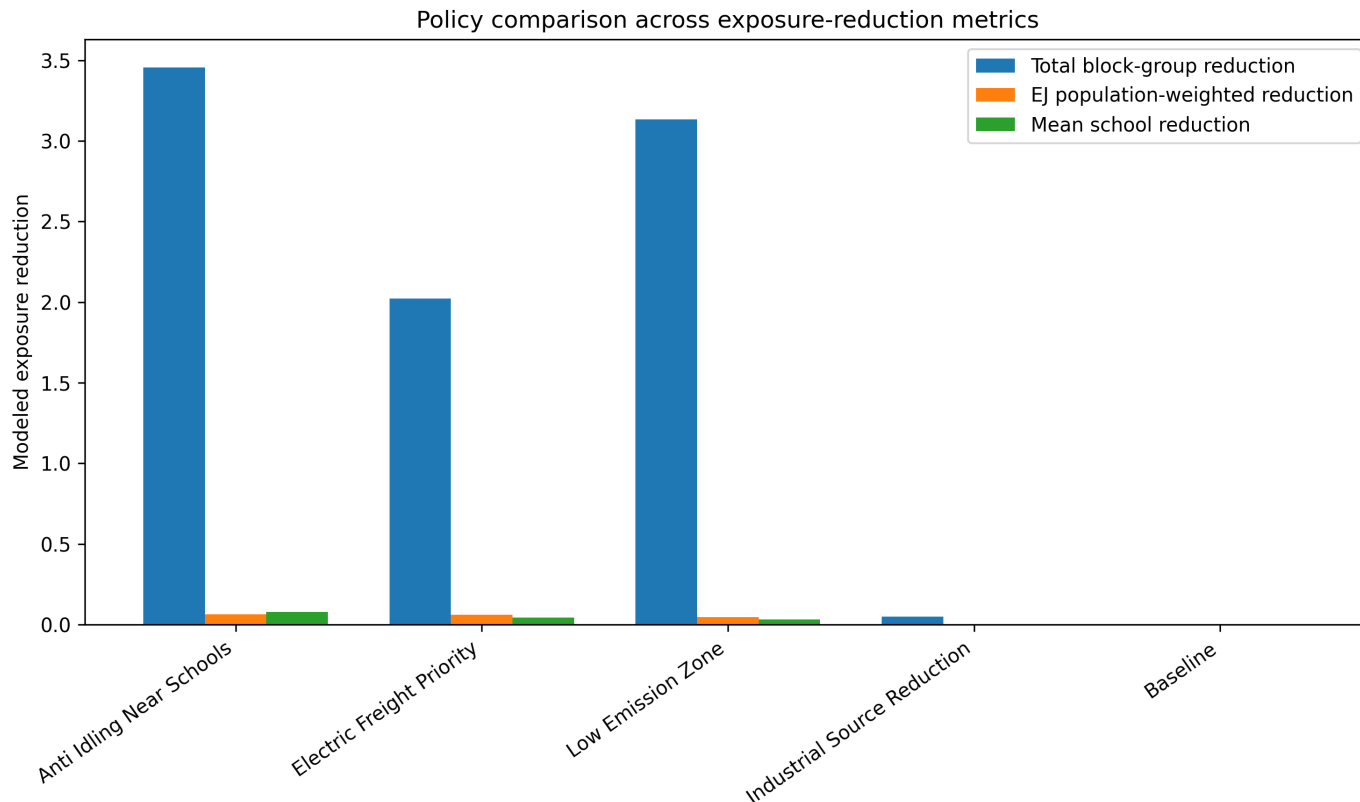


Figure 3: Policy comparison across three metrics. Anti-idling near schools produces the largest environmental justice exposure reduction in this run. Electric freight priority is a close second on the equity metric, while low-emission zones reduce total block-group exposure substantially but are less targeted to EJ exposure.

The top-ranked policy is anti-idling near schools. It reduces total block-group exposure by 3.457, environmental justice exposure by 0.063, and mean school exposure by 0.080. This makes intuitive sense because the policy is targeted to schools, which are often embedded in residential neighborhoods. If school-adjacent road segments overlap with high-exposure environmental justice block groups, then the policy can reduce both child exposure and neighborhood exposure. The result does not mean anti-idling is always more powerful than freight electrification; rather, it means that in this particular graph, the roads near schools are also important for environmental justice exposure.

Electric freight priority is second by the environmental justice metric, reducing total block-group exposure by 2.022, environmental justice exposure by 0.062, and school exposure by 0.044. Its EJ reduction is very close to the anti-idling scenario, which suggests that freight-focused policies remain important. The reason the policy does not rank first is that it targets the highest-source road nodes, not necessarily the source nodes closest to the highest-exposure environmental justice receptors. It is possible for a road to have high source strength but to be less directly connected to the most vulnerable block groups or schools.

The low-emission zone ranks third by environmental justice benefit. It reduces total block-group exposure by 3.135, which is close to the anti-idling total reduction, but it produces a smaller EJ reduction of 0.047. This is a useful result because it shows the difference between total effectiveness and equity-targeted effectiveness. A broad central low-emission zone may reduce exposure over a large area, but if its geography does not match the most vulnerable receptors, it may not be the best equity intervention.

Industrial source reduction has the smallest effect in the current run. It reduces total block-group exposure

by 0.050, environmental justice exposure by 0.001, and school exposure by 0.001. This result should not be interpreted as proof that industrial sources are unimportant. It is primarily a consequence of the current facility layer. Only five facility nodes were included, and facility source strength was smaller than the combined road-source network. A future model with detailed EPA NEI facility records, pollutant-specific emissions, rail activity, marine terminal activity, warehouse truck trips, and industrial process data could produce a larger industrial-source effect.

Figure 4 shows the baseline exposure distribution across block groups. The highest-exposure block group has an exposure score of about 2.996, while the lower-exposure block groups are closer to 0.44–0.60. Environmental justice block groups are more common near the upper end of the exposure ranking, which explains why the environmental justice population-weighted baseline exposure is higher than the non-environmental-justice baseline.

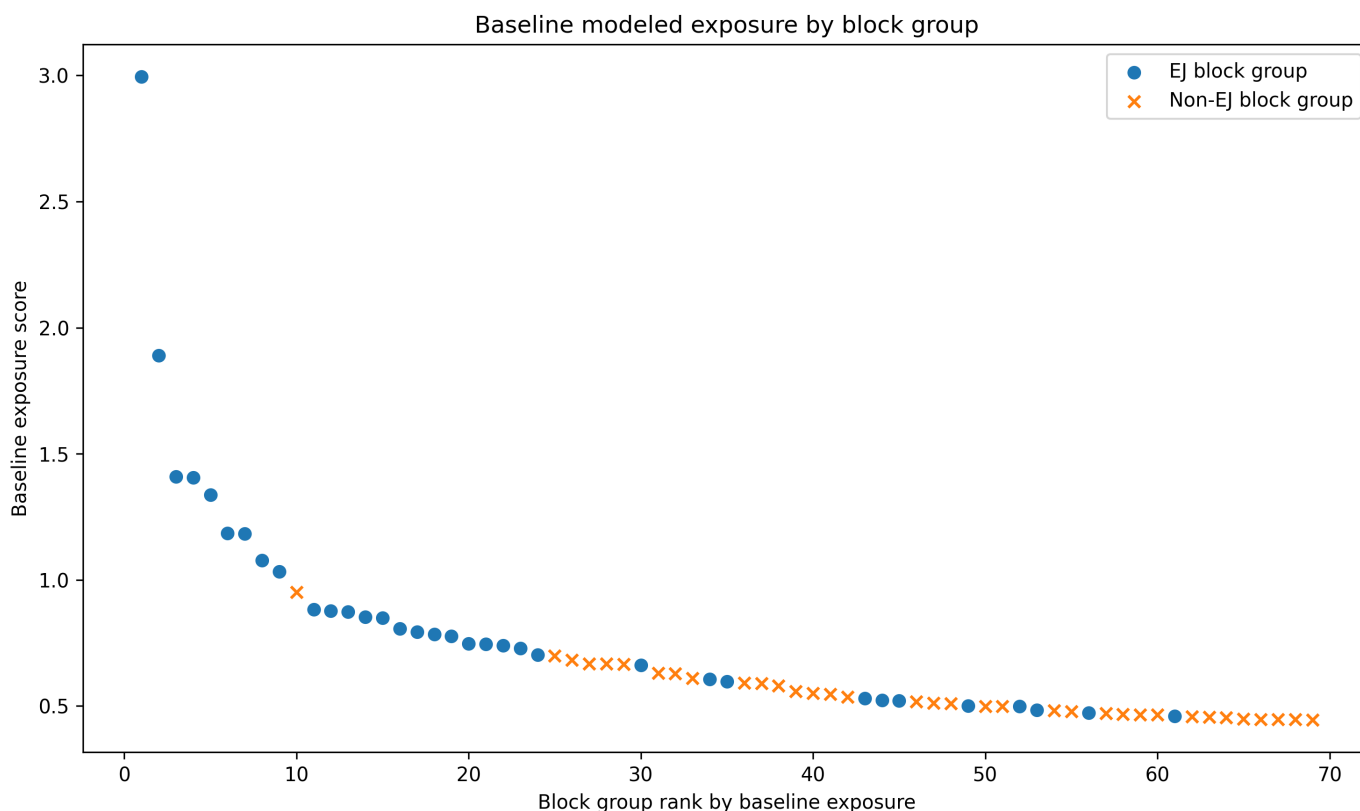


Figure 4: Baseline modeled exposure by block group. Each point is a block group ranked by exposure score. The figure shows that the highest exposure values are concentrated in a smaller subset of block groups.

Figure 5 shows the histogram of block-group exposure. Most block groups have moderate exposure scores, while a smaller number of block groups have substantially higher modeled exposure. This right-skewed pattern is typical of network exposure problems: a few locations can be much closer to multiple high-source nodes or can sit in highly connected parts of the spatial graph.

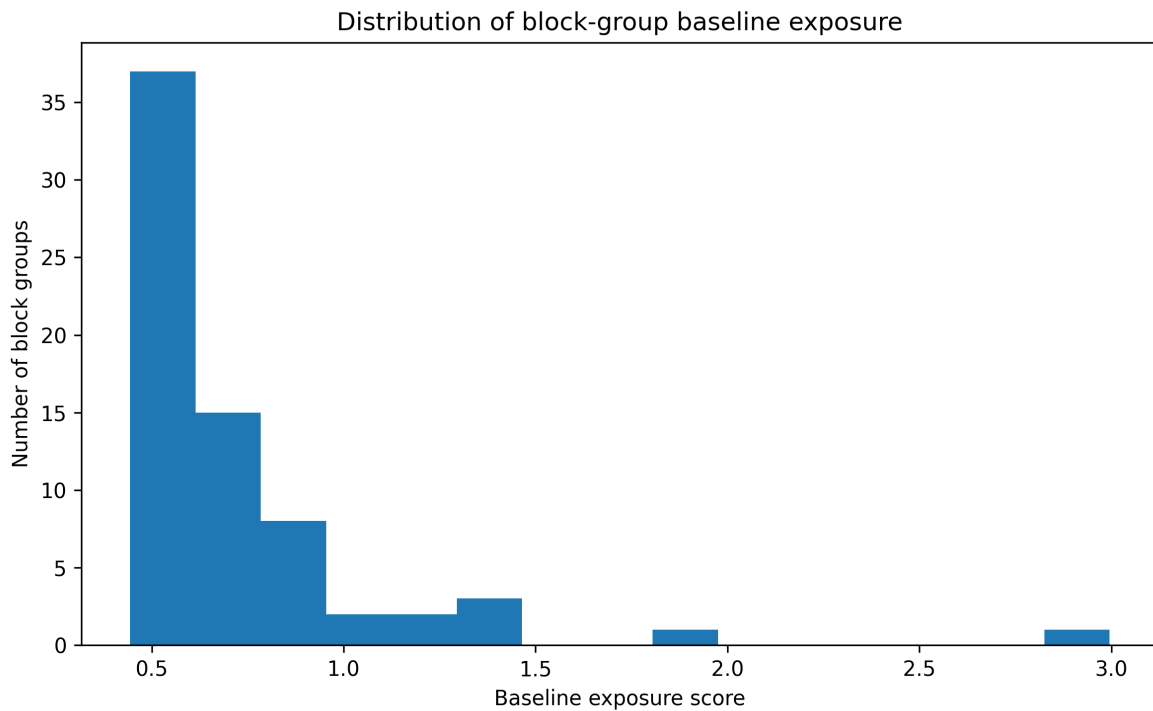


Figure 5: Distribution of baseline block-group exposure scores. The model identifies a smaller number of block groups with higher relative exposure.

Figure 6 ranks the top school receptor nodes by baseline exposure. The highest school exposure score is 3.276. The anti-idling scenario performs well on the school metric because it explicitly targets road nodes close to schools.

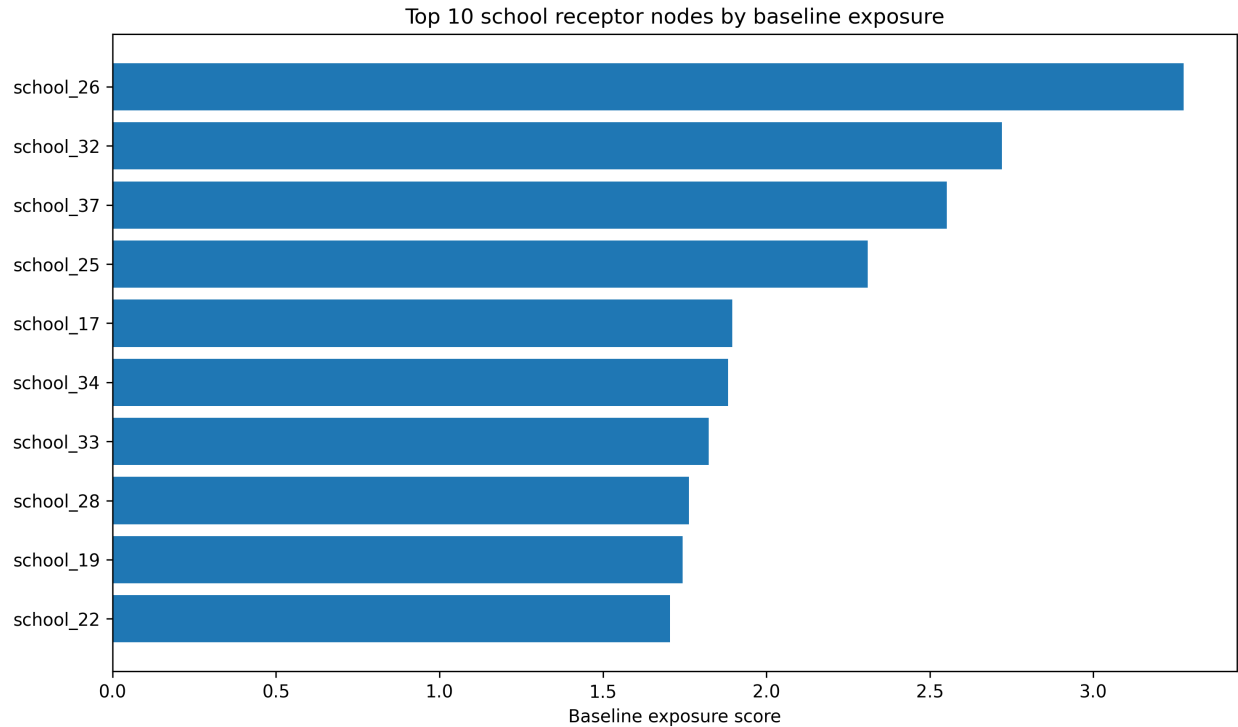


Figure 6: Top 10 school receptor nodes by baseline exposure. These nodes identify where school-adjacent traffic controls could be most important.

Figure 7 ranks the strongest source nodes, showing that the highest source nodes are roads rather than facilities and explaining why road-based policies dominate the policy comparison. This result is consistent with the current data design: the graph includes 370 road nodes and only 5 facility nodes.

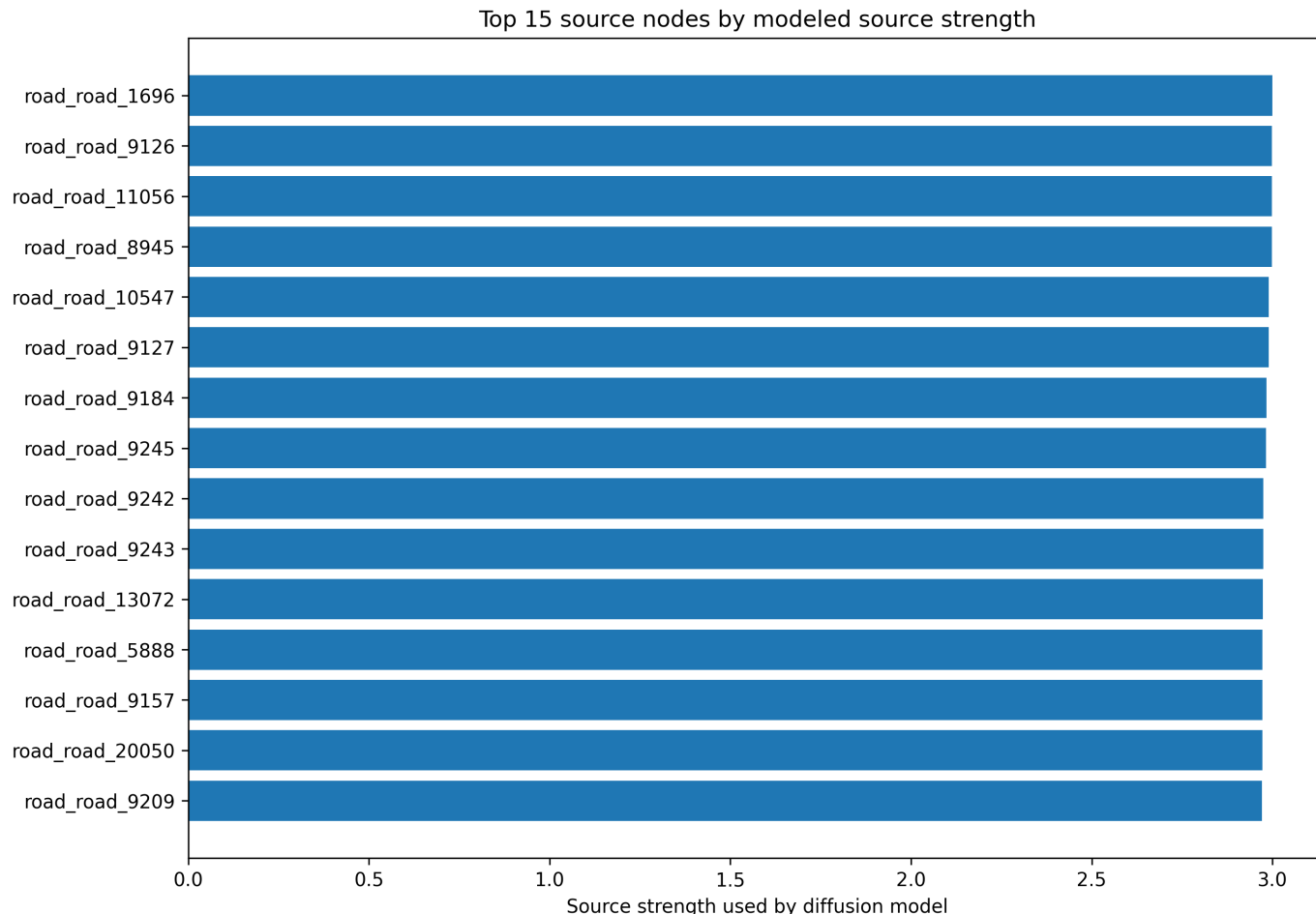


Figure 7: Top source nodes by source strength. Road nodes dominate the highest-source list in the current model, making road-based interventions more influential than industrial-source reduction.

11 Sensitivity Analysis

The baseline model uses three important graph parameters: diffusion strength $\alpha = 0.65$, connection radius $r = 1500$ meters, and decay distance $\lambda = 500$ meters. Because these choices affect how far source influence can spread through the graph, a sensitivity analysis was performed by rerunning the policy comparison over several parameter combinations. The purpose of the sensitivity analysis was not to find a single perfect parameter value. Instead, it tested whether the main policy conclusion changes when diffusion is weaker or stronger and when the spatial graph is more local or more connected.

The tested diffusion strengths were $\alpha = 0.45, 0.55, 0.65, 0.75$. Higher α means that exposure propagates farther through the graph before dissipating. The tested connection radii were 1000, 1500, and 2000 meters, and the tested decay distances were 300, 500, and 800 meters. Smaller radii and decay distances create a more local model in which sources mainly affect nearby receptors. Larger values create a more regional model in which source reductions can affect a wider part of the network.

Figure 8 shows environmental justice exposure reduction as α changes while holding radius at 1500 meters and decay distance at 500 meters. All non-baseline policies become stronger as α increases, which is expected: if influence travels farther through the graph, then reducing a source node affects more receptors. Anti-idling near schools ranks first at $\alpha = 0.45, 0.55, 0.65$, while electric freight priority becomes first at $\alpha = 0.75$.

This means the main recommendation is partly scale-dependent. In a more localized exposure interpretation, anti-idling near schools is strongest. In a more network-wide exposure interpretation, electric freight priority gains importance because high-source roads can influence a larger set of receptors.

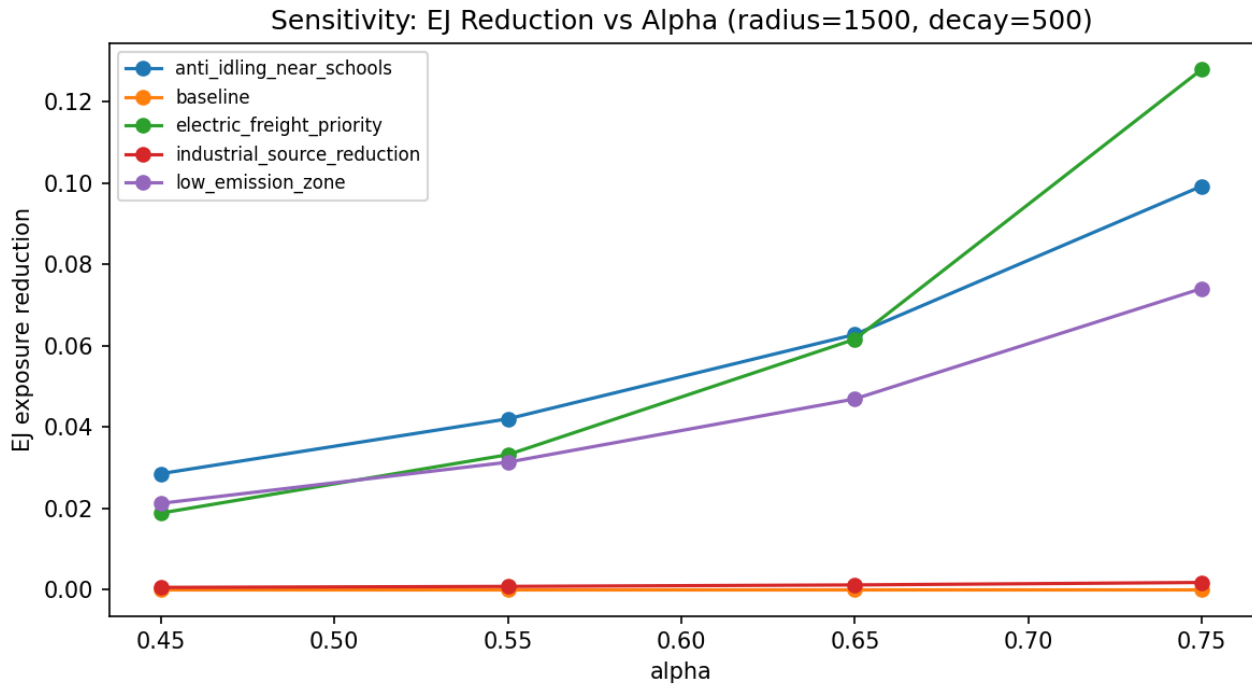


Figure 8: Sensitivity of environmental justice exposure reduction to diffusion strength α , with radius fixed at 1500 meters and decay fixed at 500 meters. Anti-idling near schools is strongest at the baseline value $\alpha = 0.65$, while electric freight priority becomes strongest under the largest tested diffusion strength.

Table 3 reports the same fixed-radius, fixed-decay sensitivity numerically. The key observation is that the top three policies remain the same road-based interventions across all α values, while industrial source reduction remains much smaller. The exact ordering of anti-idling and electric freight changes only when diffusion becomes strongest.

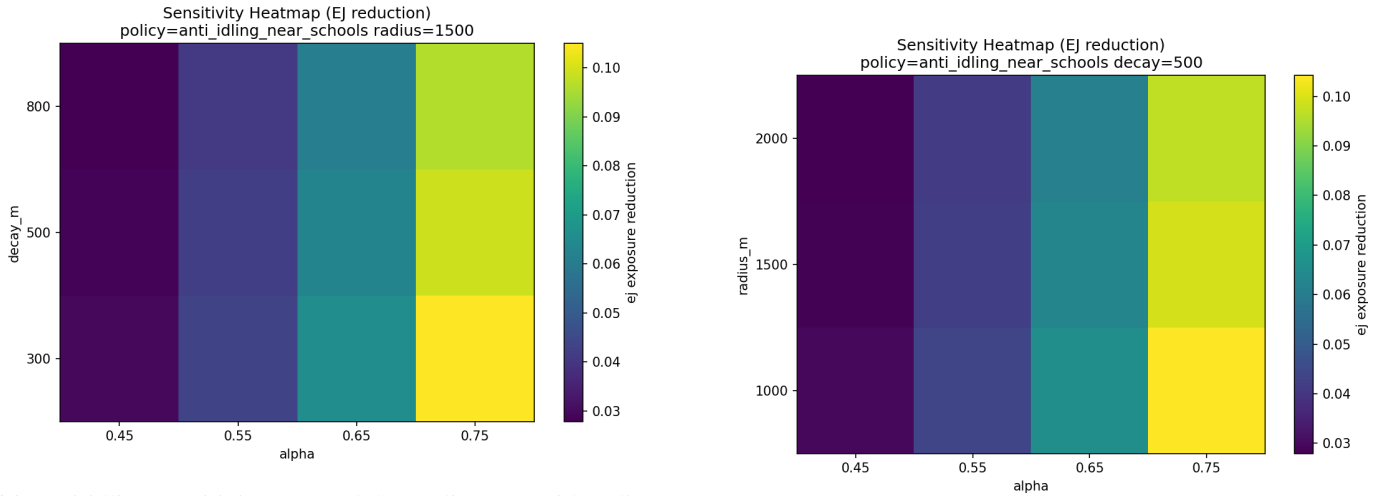
Table 3: Environmental justice exposure reduction as diffusion strength changes, with radius = 1500 meters and decay = 500 meters.

Policy	$\alpha = 0.45$	$\alpha = 0.55$	$\alpha = 0.65$	$\alpha = 0.75$
Anti-Idling Near Schools	0.0286	0.0420	0.0628	0.0992
Electric Freight Priority	0.0189	0.0332	0.0616	0.1279
Low-Emission Zone	0.0213	0.0314	0.0469	0.0741
Industrial Source Reduction	0.0006	0.0008	0.0012	0.0018
Baseline	0.0000	0.0000	0.0000	0.0000

The heatmaps in Figures 9a and 10b show how the spatial parameters affect two leading policies. For anti-idling, the strongest values occur when α is high. The effect is present across decay distances because the

policy is already spatially targeted to schools. Changing the decay distance changes how broadly that local reduction propagates, but the policy remains effective whenever diffusion is strong enough to carry the benefit into nearby environmental justice receptors.

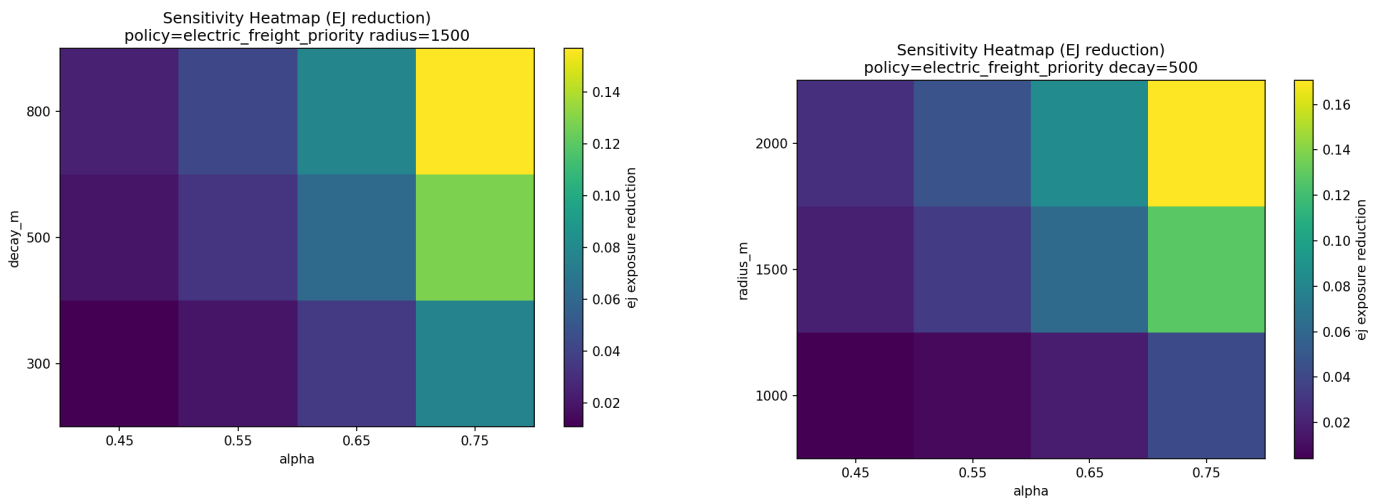
For electric freight priority, the sensitivity to radius is more pronounced. The policy targets high-source road nodes; if the graph radius is small, those roads affect only their closest receptors. If the radius is larger, the same high-source reductions can reach more block groups, so the EJ benefit grows. This explains why electric freight priority can overtake anti-idling under broader-diffusion assumptions.



(a) Anti-idling sensitivity to α and decay distance, with radius fixed.

(b) Anti-idling sensitivity to α and radius, with decay fixed.

Figure 9: Sensitivity heatmaps for anti-idling near schools. The policy remains strong across tested settings, especially when diffusion strength is larger.



(a) Electric freight sensitivity to α and decay distance, with radius fixed.

(b) Electric freight sensitivity to α and radius, with decay fixed.

Figure 10: Sensitivity heatmaps for electric freight priority. The policy benefits most when the graph allows high-source road reductions to influence a wider set of environmental justice receptors.

The sensitivity analysis strengthens the interpretation of the results in two ways. First, it shows that the

broad recommendation is robust: road-based interventions consistently dominate industrial source reduction in the current dataset. Second, it reveals that the choice between anti-idling and electric freight depends on the spatial scale of exposure. If decision-makers care most about immediate local exposure around schools, anti-idling is the best first action. If they care about a wider network of freight corridors and neighborhood exposure, electric freight priority may be equally important or even stronger. Therefore, the final policy recommendation should not choose only one. The city should treat anti-idling as a near-term targeted intervention and electric freight priority as a longer-term structural intervention.

12 Why the Results Make Sense

The results make sense for three reasons. First, the model is road-heavy because there are 370 road nodes and only 5 facility nodes. Since roads represent most sources in the graph, policies that reduce road-source strength have larger effects. This explains why anti-idling, electric freight, and low-emission zones all outperform industrial source reduction.

Second, the winning policy is not simply the largest road reduction. Anti-idling near schools reduces a smaller subset of road nodes, but those roads are spatially important because they are near school nodes and, in the current graph, also connected to environmental justice receptors. That makes the policy efficient from an equity perspective. The model therefore captures a key idea in environmental justice planning: the best intervention is not always the one that reduces the largest amount of pollution overall; it is the one that reduces pollution where vulnerable people are most exposed.

Third, electric freight priority is still very strong because it is only slightly behind anti-idling on environmental justice reduction. This is important because freight electrification is a more structural intervention. Anti-idling enforcement can be implemented quickly with signs, enforcement, education, school-zone rules, and curb-management changes, while electric freight requires vehicles, charging infrastructure, fleet turnover, incentives, and coordination with logistics operators. The model suggests that both strategies are valuable, but they operate on different time horizons.

The low-emission zone result is also meaningful because it reduces total exposure more than electric freight priority, but it ranks lower on environmental justice benefit. This means the zone is useful, but not maximally targeted. A city should not assume that a central low-emission zone automatically produces the best equity result because the zone boundary matters. A future optimization model could search for the best low-emission-zone boundary rather than using a fixed center-radius approach.

Industrial source reduction is small in this run, but the interpretation should be careful because the facility layer was simplified. The EPA NEI framework indicates that point sources can include large industrial facilities, power plants, airports, and other industrial or commercial sources (U.S. Environmental Protection Agency, 2026a). A more detailed facility dataset could change the ranking, especially if high-emitting sources are close to residential areas or schools. Therefore, the current result should motivate better facility data collection, not dismissal of stationary-source policies.

13 Recommendations to the City

Based on the current model, the first recommendation is to prioritize anti-idling enforcement near schools and dense residential areas. This policy ranked first for environmental justice exposure reduction and also produced the largest school exposure reduction. A practical city program could begin with the highest-exposure school nodes and nearby road segments. The city could install anti-idling signs, increase enforcement during pickup and dismissal periods, coordinate with school transportation providers, create truck no-idling zones near school entrances, and monitor compliance using periodic observations. Because this intervention is relatively

low-cost and does not require major infrastructure, it is a strong near-term recommendation.

The second recommendation is to develop an electric freight priority program for the highest-source road corridors. The electric freight scenario ranked second by environmental justice benefit and was very close to anti-idling. A city cannot electrify every freight movement immediately, so the model should be used to identify the top road nodes by source strength and receptor connection. Those corridors should be prioritized for charging infrastructure, fleet incentives, zero-emission delivery pilots, and partnerships with freight operators. If local agencies or state partners can identify routes with high truck activity, those routes should be overlaid with school and environmental justice receptors before investments are made.

The third recommendation is to redesign the low-emission-zone idea around equity rather than geography alone. The tested low-emission zone reduced total exposure, but it was not the best equity policy. Instead of choosing a zone simply because it is central, the city should define a low-emission priority area by combining source intensity, EJ population, school proximity, and baseline exposure. In future work, this can be formulated as an optimization problem: choose the set of road or facility nodes to regulate so that environmental justice benefit is maximized under a budget constraint.

The fourth recommendation is to improve industrial and freight-facility data. The current facility layer is too small for strong conclusions. The city should use EPA NEI point-source data, state permitting records, local parcel data, warehouse/loading-zone data, and truck-trip observations to build a more complete source inventory. This will help separate emissions from roads, industrial operations, freight terminals, rail, marine activity, and loading zones. Without that separation, the model may understate non-road sources.

The fifth recommendation is to validate the model with air-quality sensors. The iSUPER sensor network measures particulate matter and gases such as NO_x, ozone, and carbon monoxide, with some sensors measuring wind speed and wind direction (iSUPER, 2026). Those data could be used to compare predicted exposure hotspots with observed pollution patterns. Even a simple validation step, such as correlating sensor PM or NO_x averages with nearby modeled exposure scores, would make the model much stronger.

The sixth recommendation is to create an iterative city dashboard because the graph model is not useful only once; it can be updated as better data arrive. Traffic counts, truck percentages, school locations, new facilities, construction changes, and sensor readings can all be added. Each update can rerun the same intervention scenarios and show whether priorities changed. A transparent model can support public meetings because residents can see why certain roads or schools were prioritized.

14 Strengths and Weaknesses

A useful way to evaluate the project is to separate what the model does well from what it cannot yet do. This is especially important because the model is intentionally between two extremes. It is more detailed than a purely qualitative policy discussion, but it is not as detailed as a full regulatory air-quality model. Its value is in transparent prioritization.

14.1 Strengths

The first strength is transparency. Every major step is visible: nodes are created from block groups, roads, schools, and facilities; edges are created by distance; source strengths are assigned to roads and facilities; exposure is computed with a diffusion equation; and policy scenarios change the source vector. This makes the model easier to explain to a nontechnical audience than a black-box machine-learning model. A city planner, teacher, or community member can understand why a school near several strong source nodes receives a higher modeled exposure score.

The second strength is modularity. The same graph structure can accept better data later. If MassDOT traffic counts replace the imputed AADT values, the source vector improves without changing the diffusion

equation. If official MassGIS environmental justice attributes replace the current binary approximation, the evaluation metric improves without changing the edge-building process. If sensor data are added, they can be used to validate predictions without discarding the existing workflow.

The third strength is policy relevance. The model does not stop at identifying high-exposure areas. It directly compares interventions. This matters because cities face budget, staffing, and political constraints. A map of exposure is useful, but a ranking of interventions is more actionable. The model can say that, under current assumptions, anti-idling near schools slightly outperforms electric freight priority on EJ exposure reduction, while the sensitivity analysis shows that electric freight becomes stronger at broader diffusion scales.

The fourth strength is its environmental justice focus. Policies are ranked by environmental justice exposure reduction rather than by total exposure reduction alone. This prevents a policy from appearing best simply because it reduces exposure in less vulnerable or less populated areas. The model therefore aligns the mathematics with the ethical goal of reducing unequal burdens.

14.2 Weaknesses

The first weakness is input uncertainty. Several important values are estimated, including road-level AADT, truck share, facility emission scores, and the simplified EJ flag. These estimates make the model usable, but they also limit the strength of any real-world conclusion. The results should therefore be presented as a demonstration of a method and an initial policy screen, not as proof that one city policy will definitely outperform another.

The second weakness is that the exposure score is unitless. It cannot be compared to air-quality standards, health thresholds, or measured pollutant concentrations. This means the model can rank policies within the project, but it cannot say whether a school exceeds a PM_{2.5} standard or whether an intervention reduces asthma risk by a specific amount.

The third weakness is the absence of meteorology and time. Pollution is strongly shaped by wind, temperature, atmospheric mixing, congestion, and time-of-day traffic patterns. The current graph is static. It cannot distinguish a truck corridor that matters mostly during morning delivery from one that matters all day. It also cannot represent downwind and upwind differences. This limits the model's physical realism.

The fourth weakness is the small facility layer. Because only five facilities are included and their source scores are relative, the model likely underestimates stationary and industrial contributions. The weak performance of industrial source reduction should therefore be interpreted as a property of the current dataset, not as a conclusion that industrial sources are unimportant.

The fifth weakness is that costs and feasibility are missing. Anti-idling enforcement, freight electrification, low-emission zones, and industrial retrofits have very different implementation costs. A city may prefer a policy with slightly lower exposure reduction if it is much cheaper, easier to enforce, or more politically feasible. A complete decision model should add cost-effectiveness and implementation difficulty.

Key Result

Overall, the model's strongest use is as a transparent early-stage decision-support tool. It is good at comparing policies under shared assumptions, but it should be paired with better traffic counts, official EJ attributes, facility emissions, sensor validation, and cost estimates before being used for final city planning decisions.

15 Limitations

The most important limitation is that several key values are estimated. The traffic data used in the final run are expected values generated from a gravity-style proxy, not official traffic counts for every road segment.

The imputation was calibrated using realistic ranges from the Route 16 corridor study, but it does not replace MassDOT or CTPS road-level count data. The final model should use measured AADT and truck percentages wherever available.

A second limitation is the environmental justice label. The model now contains both EJ and non-EJ groups, which is necessary for comparison, but the current split should be treated as an approximate modeling field. A final study should replace this with the official MassGIS EJ attributes and should preserve the specific criteria behind the designation, such as income, minority population, English isolation, or combinations of these indicators.

A third limitation is that the model is static, even though pollution varies by hour, day, season, weather, fleet composition, congestion, and operational activity. Trucks during the morning peak may affect different receptors than trucks late at night, and wind direction can shift exposure from one side of a road to another. The current model uses one static graph and one source vector, while a better version would use a temporal graph with separate morning, afternoon, evening, and weekend scenarios.

A fourth limitation is that the model does not represent physical atmospheric dispersion. It uses distance-decay graph diffusion, not computational fluid dynamics or Gaussian plume modeling. This is appropriate for a first-pass policy model, but it means the exposure scores are relative indices. They should not be interpreted as PM_{2.5} concentration, NO₂ concentration, cancer risk, or respiratory hazard index.

A fifth limitation is the simplified facility layer because only five facility nodes were included, and facility source scores were relative. EPA NEI data can provide more detailed facility information, but the current run has not fully incorporated pollutant-specific facility emissions. This likely explains why industrial source reduction has a small modeled effect.

A sixth limitation is that costs are not included. A policy that produces the largest exposure reduction may not be the most cost-effective. Anti-idling enforcement, freight electrification, low-emission zones, and industrial retrofits have very different costs and implementation barriers. A future model should add cost estimates and compute exposure reduction per dollar.

16 Future Work

The next version of the project should replace imputed traffic values with measured counts. MassDOT traffic counts, CTPS count locations, and local traffic studies can be joined to road segments. Where direct counts are missing, a better gravity model can be fitted using road functional class, distance to highways, proximity to industrial parcels, school zones, and known count stations. This would produce a more defensible source-strength layer.

The second improvement is to incorporate wind, which can be represented as a directional edge multiplier:

$$w_{ij}^{\text{wind}} = w_{ij}(1 + \gamma A_{ij}),$$

where A_{ij} is a wind-alignment factor, downwind edges would become stronger, and upwind edges would become weaker. Sensor or NOAA wind data could be used to create seasonal scenarios.

The third improvement is to optimize interventions rather than test fixed scenarios. For example, instead of reducing the top 20 percent of road nodes, the model could choose the best k road nodes to reduce:

$$\max_{S: |S| \leq k} \sum_{i \in B_{\text{EJ}}} P_i(x_i^{\text{baseline}} - x_i^S).$$

This formulation would turn the project from scenario comparison into a true policy optimization model that

searches for the best set of interventions rather than evaluating only manually chosen scenarios.

The fourth improvement is validation with sensor observations. If PM, NO_x, or black carbon data are available, the model can be calibrated so that predicted exposure aligns with measured values. If not, the model can still be used as a screening tool, but conclusions should remain relative.

17 Conclusion

This paper developed a graph-based diffusion model for environmental justice pollution exposure in Chelsea and Everett, Massachusetts. The model represents roads, block groups, schools, and facilities as a spatial influence network. Road and facility sources are assigned relative source strengths, exposure diffuses through the graph, and policies are compared by their ability to reduce total exposure, environmental justice exposure, and school exposure.

The final run used 486 nodes and 20,303 edges. The baseline model found higher population-weighted exposure in the environmental justice group than in the non-environmental-justice comparison group. Among the tested interventions, anti-idling enforcement near schools produced the largest environmental justice exposure reduction in the baseline run, narrowly ahead of electric freight priority. The sensitivity analysis showed that this ordering is not absolute: electric freight priority becomes strongest when diffusion is allowed to travel farther through the graph. Low-emission zones reduced total exposure but were less targeted to EJ benefit, while industrial source reduction had a small effect because the current facility layer was limited.

The central recommendation is that Chelsea and Everett should begin with targeted anti-idling controls near schools and high-exposure residential areas, while simultaneously building a longer-term freight electrification strategy for high-source corridors. The city should also improve facility-source data, validate model outputs with sensor observations, and eventually convert the scenario model into an optimization tool. The model is not a final regulatory air-quality assessment, but it is a useful mathematical framework for deciding where limited pollution-reduction resources should be targeted first.

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